Map Filter Reduce

🧠 Mental Model First (Don't Memorize Syntax)

Method	Think Like
map()	"Transform every item"
filter()	"Keep only what matches"
reduce()	"Combine everything into one result"

1️⃣ map() — Transform Data

Definition:

Runs a function on **every element** and returns a new array.

```
const prices = [100, 200, 300];

const withGST = prices.map(p => p * 1.18);
```

Output:

[118, 236, 354]

2️⃣ filter() — Select Data

Definition:

Keeps only elements where condition is true.

```
const orders = [1200, 300, 800, 1500];

const bigOrders = orders.filter(o => o > 1000);
```

Output:

[1200, 1500]

3️⃣ reduce() — Combine Into One Value

This is the most powerful (and most misunderstood).

Definition:

Reduces the entire array into a **single result**.

```
const nums = [1, 2, 3, 4];


const sum = nums.reduce((acc, curr) => acc + curr, 0);
```

Output:

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Immutability & Data Pipelines

Instead of editing original document,
you create a new version each time.

Git works like this.
So should your code.

Immutability = never modify original state.

Q.Why is immutability important in MERN apps?

Because frontend + backend share predictable state assumptions.